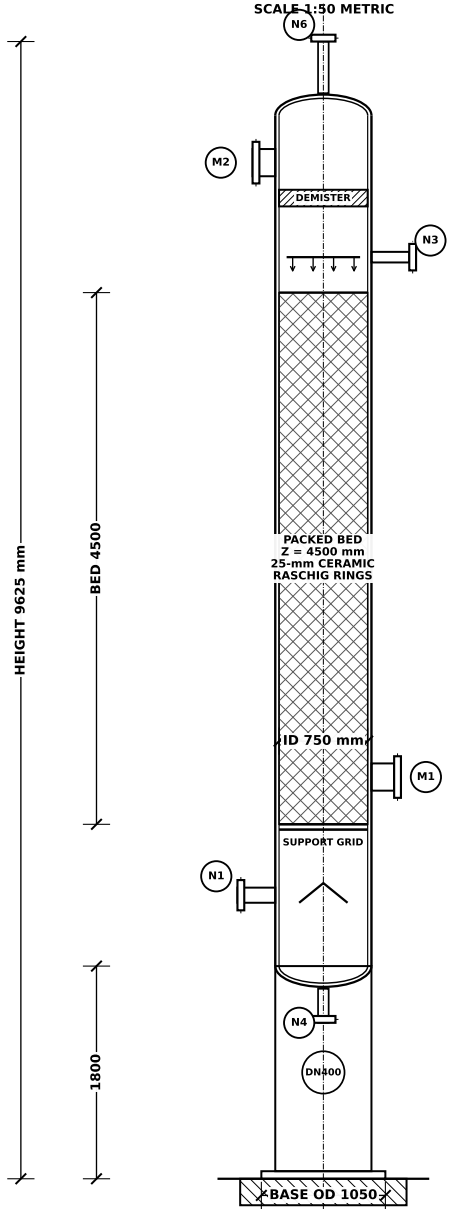
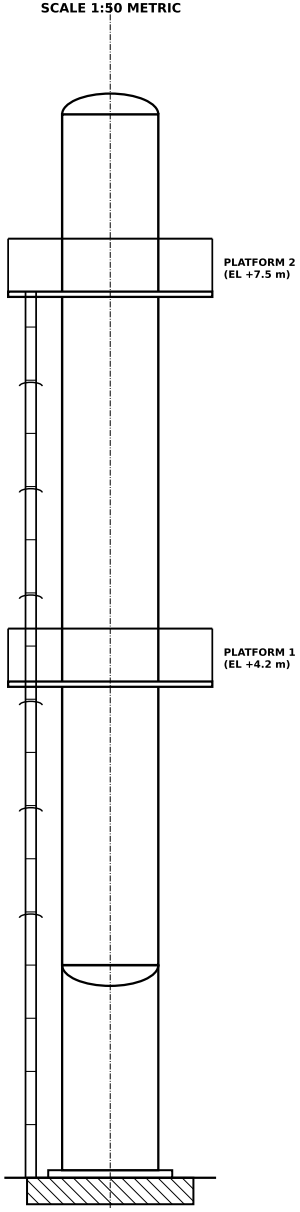
SECTIONAL ELEVATION A-A

SCALE 1:50 METRIC



EXTERIOR ELEVATION & GA

SCALE 1:50 METRIC



VESSEL MECHANICAL DESIGN & OPERATING CONDITIONS		
PARAMETER	**VALUE**	**SPECIFICATION / BASIS**
Process Service	Wet Bromine Scrubbing	Absorption of 2.0% Br2 with Water
Internal Diameter (ID)	**750 mm ID (Ac=0.44 m2)**	70% Flooding Limit (vG = 0.599 m/s)
Shell & Head Material	SA-516 Gr 70 Carbon St.	8 mm Shell / Head TBD (Prelim CAD)
Corrosion Barrier	3.0 mm Bonded PVDF	15 kV High-Voltage Spark Tested
Design Press / Temp	1.50 bar g / 65 DEG C	Full Vacuum (-1.0 bar g) Rating
Operating Press / Temp	1.07 bar a / 25 DEG C	Inlet Gas at P = 803 mmHg
Inlet Gas Flow (QG)	0.25 m3/s (900 m3/h)	Inlet: 2.0 mol% -> Outlet: 0.20 mol%
Absorbent Water (L)	17.25 kg/s (62.1 t/h)	Demineralized Water (L/G = 90.5)
Packed Bed Depth (Z)	**4500 mm (Vp = 1.79 m3)**	25-mm Glazed Ceramic Raschig Rings
Overall Column Height	9625 mm (T/T: 7200 mm)	Ground to Top Vent Flange Face
Operating / Flood Load	16.1 kN / 26.4 kN	Normal Service / Flooded Upset

NOZZLE SCHEDULE (ASME B16.5 CLASS 150 RF)				
TAG	**SERVICE DESCRIPTION**	**SIZE (DN)**	**RATING**	**FACING**
N1	Gas Inlet Mixture	DN 150 (6")	CL 150	RF Flange
N3	Water Absorbent Inlet	DN 80 (3")	CL 150	RF Flange
N4	Rich Liquor Outlet	DN 80 (3")	CL 150	RF Flange
N5	Column Shell Drain	DN 40 (1.5")	CL 150	RF Flange
N6	Treated Gas Vent	DN 50 (2")	CL 150	RF Flange
M1	Lower Packing Access	DN 450 (18")	CL 150	RF Blind
M2	Upper Internals Manway	DN 450 (18")	CL 150	RF Blind

GENERAL ENGINEERING NOTES

1. **ALL DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE.**
2. **DESIGN BASIS: PRELIMINARY MECHANICAL DESIGN— SUBJECT TO ASME SEC VIII DIV 1 VERIFICATION.**
3. **PROCESS VALUES: COLUMN ID 750 mm, PACKED BED DEPTH 4500 mm (Vp = 1.79 m3).**
   PACKING MASS 1.22 t, WATER FLOW 17.25 kg/s, DRY GRAVITY LOAD 11.9 kN.
4. PRELIMINARY CAD: SHELL THK 8 mm (PRELIM), HEAD THK TBD (PRELIM), SKIRT HT 1800 mm, BASE RING OD 1050 mm.
5. MOC: SA-516 GR 70 WITH 3.0 mm BONDED PVDF LINER FOR WET BROMINE SERVICE.
6. ANCHORAGE: 8x M24 ANCHOR BOLTS (SA-193 B7) EMBEDDED 450 mm IN CONCRETE.

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PACKED-BED ABSORBER (C-101) — GENERAL ARRANGEMENT & ELEVATION	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	1:50 METRIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	PEDD-PBA-01	29 / 09 / 2026

PACKED ABSORBER (C-101) | 0.25 m3/s GAS (2% Br2) | WATER: 17.25 kg/s | ASME SEC VIII DIV 1 / B16.5